The documentation and process conversion measures necessary to comply with this document shall be completed by 16 June 2023.

INCH-POUND

MIL-PRF-19500/398L
w/AMENDMENT 3
16 March 2023
SUPERSEDING
MIL-PRF-19500/398L
w/AMENDMENT 2
14 February 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, HIGH-FREQUENCY,
ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT) AND UNCAPSULATED, RADIATION HARDNESS ASSURANCE, DEVICE TYPE 2N3866, QUALITY LEVELS: JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN silicon 2N3866 transistors for use in VHF-UHF amplifier applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS,) are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA), eight radiation levels is provided for quality levels JANTXV, JANS, JANHC, and JANKC. RHA level designators “M”, “D”, “P“, “L”, “R”, “F”, “G”, and “H” are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Package and die outlines. The device package for the encapsulated device types are as follows: TO-205AD, formally TO-39, in accordance with figure 1 and surface mount in accordance with figure 2. The dimensions and topography for JANHC and JANKC unencapsulated die are in accordance with figure 3.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T (1) T_A = (2)	P_T T_C = (3)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}	$R_{\theta JC}$	$R_{\theta JA}$
	W	W	V dc	V dc	V dc	A dc	°C	°C/W	°C/W
2N3866	1.0	2.9	60	30	3.5	0.4	-65 to +200	60	175
2N3866UB	0.5		60	30	3.5	0.4	-65 to +200		325

(1) Derate linearly 5.71 mW/°C (2N3866) and 3.08 mW/°C (2N3866UB) above $T_A \geq$ +25°C.
(2) T_A = room ambient as defined in the general requirements of MIL-PRF-19500.
(3) P_T = 2.9 W at T_C = +25°C, derate at 16.6 mW/°C above T_C > +25°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.4 Primary electrical characteristics.

	h_{FE} (1) V_{CE} = 5.0 V dc I_C = 50 mA dc		$\lvert h_{fe} \rvert$ V_{CE} = 15 V dc I_C = 50 mA dc f = 200 MHz		$V_{CE(SAT)}$ I_C = 100 mA dc I_B = 10 mA dc	C_{obo} V_{CB} = 28 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	P_{out1} V_{CC} = 28 V dc P_{in} = 0.15 W f = 400 MHz	P_{out2} V_{CC} = 28 V dc Pin = 0.075 W f = 400 MHz
	2N3866	2N3866A	2N3866	2N3866A	V dc	pF	W	W
Min	15	25	2.5	4.0			1.0	0.5
Max	200	200	8.0	7.5	1.0	3.5	2.0	

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

1.5.1 JAN brand and quality level designators.

1.5.1.1 Encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.1.2 Unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H").

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "3866".

1.5.4 Suffix symbols. The following suffix symbols are incorporated in the PIN as applicable.

1.5.4.1 First suffix symbol. The first suffix symbol "A" indicates that the transistor is a modified version of the approved device type.

1.5.4.2 Following suffix symbols. The following suffix symbols are incorporated in the PIN for this specification sheet:

	A blank second suffix symbol indicates a through-hole mount package (see figure 1).
UB	Indicates a 4 pad surface mount package (see figure 2).
AUB	Indicates a 4 pad surface mount package (see figure 2).

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifier that is applicable for this specification sheet is "A" (see figure 3 and 6.5).

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* 1.6 Radiation features. The following radiation features are applicable for RHA devices supplied to this specification sheet.

* 1.6.1 Maximum total ionizing dose (TID). The maximum TID that RHA devices were tested to in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750 are as follows:

For device type 2N3866, 2N3866UB: ..... 300 krads(Si) 1/

* 1/ The manufacturers supplying these device types have performed characterization testing in accordance with condition A (dose rate = 50 to 300 rad(Si)/s) of method 1019 of MIL-STD-750. The radiation end point limits are guaranteed only to a maximum TID level of 300 krads(Si).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

η: ..... (eta) Collector efficiency = $\frac{\text{rf power out}}{\text{dc power in}} \times 100$

Pin: Input power
Pout: Output power

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3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and herein. The device package styles shall be as follows: Three pin metal can (TO-205AD, formally TO-39) in accordance with figure 1, four pad surface mount case outline UB in accordance with figure 2, and unencapsulated die in accordance with figure 3 for device types JANHC and JANKC.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements and test levels shall be as defined in MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Marking.

3.7.1 Through hole mount packages. Marking shall be in accordance with MIL-PRF-19500.

3.7.2 Surface mount packages. Marking shall be in accordance with MIL-PRF-19500. The marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturer's symbol and logo. The prefixes JAN, JANTX, JANTXV and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted. The radiation hardened designator shall immediately precede (or replace) the device "2N" identifier (depending upon degree of abbreviation required).

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

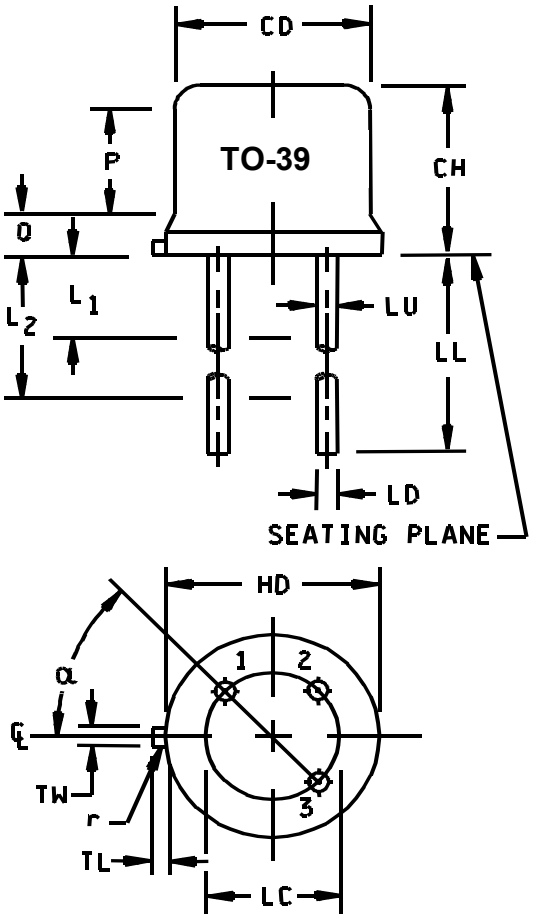
4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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Symbol	Dimensions				Note
	in	in	mm	mm	
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	6
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200	TP	5.08	TP	7
LD	.016	.021	0.41	0.53	8,9
LL	.500	.750	12.7	19.05	
LU	.016	.019	0.41	0.48	8,9
L_1		.050		1.27	8,9
L_2	.250		6.35		8,9
P	.100		2.54		7
Q		.030		0.76	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45°	TP	45°	TP	7

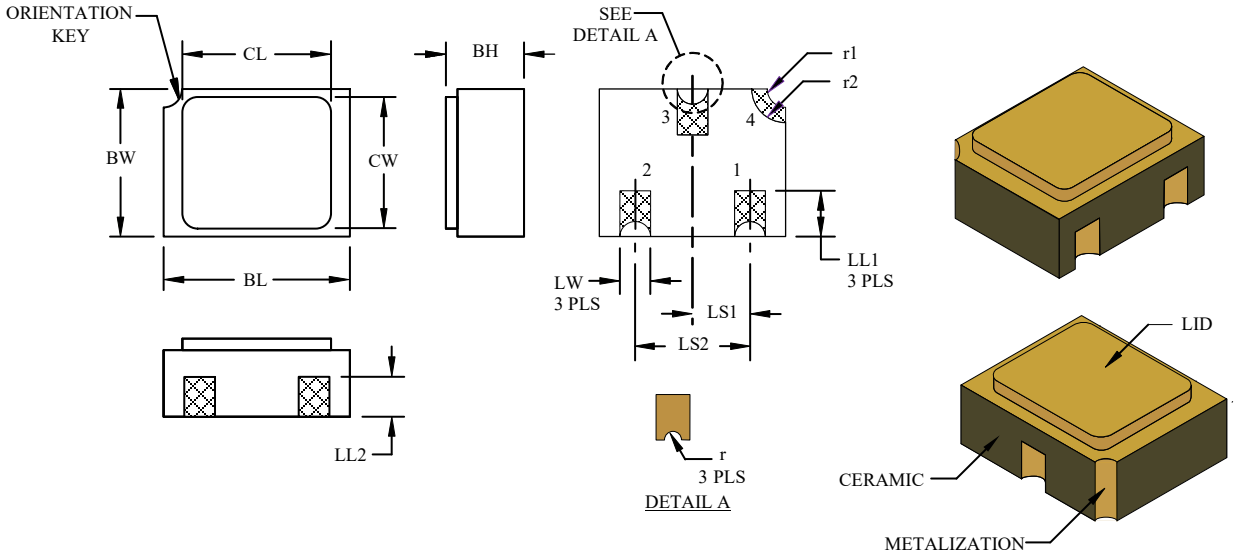


NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by gauging procedure.
8. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in and beyond LL minimum.
9. Dimension applies to all three leads.
10. The collector shall be internally connected to the case.
11. Dimension r (radius) applies to both inside corners of tab.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (TO-205AD, formally TO-39).

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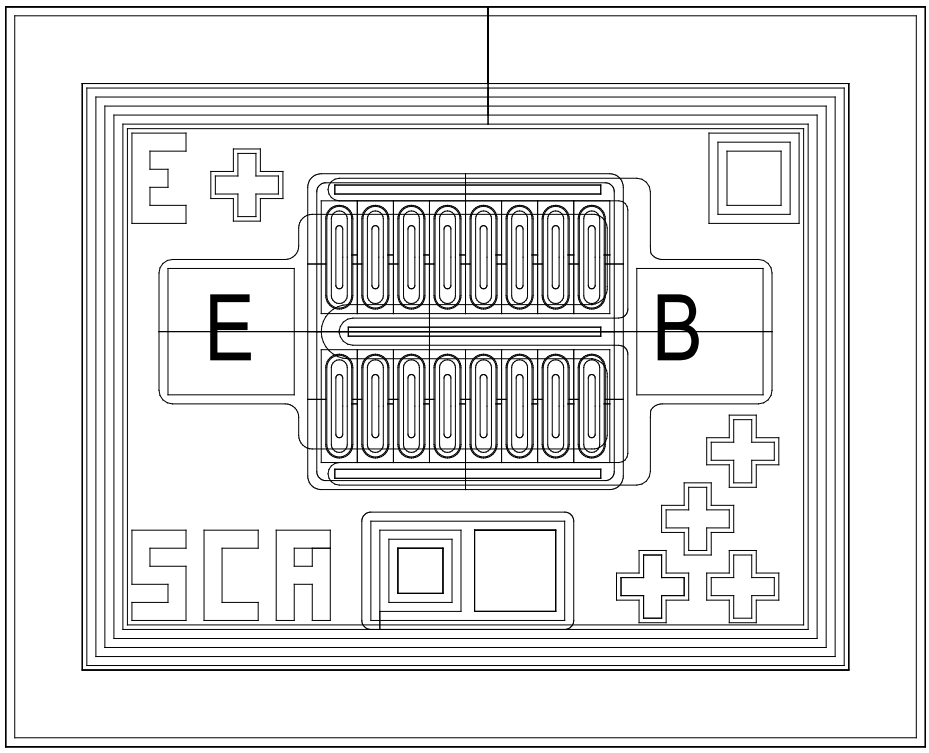


Ltr.	Dimensions				Note	Ltr.	Dimensions				Note
	in	in	mm	mm			in	in	mm	mm	
	Min	Max	Min	Max			Min	Max	Min	Max	
BH	.046	.056	1.17	1.42		LS1	.035	.040	0.89	1.02	
BL	.115	.128	2.92	3.25		LS2	.071	.079	1.80	2.01	
BW	.085	.108	2.16	2.74		LW	.016	.024	0.41	0.61	
CL		.128		3.25		r		.008		0.20	
CW		.108		2.74		r1		.012		0.31	
LL1	.022	.038	0.56	0.97		r2		.022		0.56	
LL2	.017	.035	0.43	0.89							

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.

* FIGURE 2. Physical dimensions, surface mount (UB).

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Die size:	.016 x .020 inch (0.41 x 0.51 mm).
Die thickness:	.008 ±.0016 inch (0.20 ±0.041 mm).
Base pad:	.0028 x .0028 inch (0.07 x 0.07 mm).
Emitter pad:	.0028 x .0028 inch (0.07 x 0.07 mm).
Back metal:	Gold, 6,500 ±1,950 Å.
Top metal:	Aluminum, 17,500 ±2,500 Å.
Back side:	Collector.
Glassivation:	SiO_2, 7,500 ±1,500 Å.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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4.3 Screening.

4.3.1 Screening of packaged devices (quality levels JANS, JANTX and JANTXV only). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.4).	Thermal impedance, method 3131 of MIL-STD-750 (see 4.3.4).
9	I_{CEO} and h_{FE1}	Not applicable.
11	I_{CEO} and h_{FE1}; ΔI_{CEO} = 100 percent of initial value or 2 µA dc, whichever is greater; Δh_{FE1} = ±20 percent of initial value.	I_{CEO} and h_{FE1}
12	See 4.3.2.	See 4.3.2.
13	Subgroups 2 and 3 of table I herein. ΔI_{CEO} = 100 percent of initial value or 2 µA dc, whichever is greater; Δh_{FE1} = ±20 percent of initial value.	Subgroup 2 of table I herein. ΔI_{CEO} = 100 percent of initial value or 2 µA dc, whichever is greater; Δh_{FE1} = ±20 percent of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a; TX and TXV levels do not need to be repeated in screening requirements.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750; V_{CB} = 10 to 30 V dc. Power shall be applied to achieve a junction temperature T_J = +135°C minimum and power dissipation of $P_T \geq 75$ percent of max rated P_T as defined in 1.3 herein. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.3 Screening of unencapsulated die (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500, "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.4 Thermal impedance. The thermal impedance shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining V_H, V_{CE}, I_M, I_H, t_H, and t_{MD}. The limit used in screen 3c and table I, subgroup 2 shall be set statistically by the supplier over several die lots and submitted to the qualifying activity for approval.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection.

4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B4	1037	V_{CB} = 10 V dc; 2,000 cycles, adjust power or current to achieve a ΔT_J = +100°C.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with table E-VIa of MIL-PRF-19500, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjusted T_A or P_D to achieve a T_J = +225°C minimum.

4.4.2.2 Quality level (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI (conformance inspection) shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot and/or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode. Endpoints shall be after each step and shall be in accordance with MIL-PRF-19500.

Step	Method	Conditions
1	1026	Steady-state life: 1,000 hours, V_{CB} = 10 to 30 V dc; power shall be applied to achieve T_J = +150°C minimum and a power dissipation of $P_D \geq$ 75 percent of max rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours; T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements.

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2 conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) in accordance with MIL-PRF-19500.

4.4.3.1 Quality level JANS, table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E (not applicable to UB suffix devices).
C5	3131	See 4.5.2; n = 22, c =0.
C6	1026	1,000 hours, V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a power dissipation of $P_D \geq 75$ percent of max rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Quality levels JAN, JANTX, and JANTXV, table E-VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Test condition E (not applicable to UB suffix devices).
C6		Not applicable.
C8	3005	Pre-pulse condition V_{CE} = 0, I_C = 0; pulse condition I_C = 400 mA dc, t_P = 60 s, 1 cycle; $t_r \leq 6s$, $t_f \leq 6s$. Sample size, n = 22, c = 0 (see 4.5.4).

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750 for total ionizing dose, or method 1017 of MIL-STD-750 for neutron fluence, as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Group D inspection may also be performed ahead of the screening lot using die selected in accordance with MIL-PRF-19500 and related documents. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta requirements shall be in accordance with table I, subgroup 2 herein.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power application shall be 79 mA dc minimum.

b. Collector to emitter voltage magnitude shall be 20 V dc minimum.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be $+25°C \leq T_R \leq +75°C$ and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit of $R_{\theta JC}$ shall be 60°C/W for 2N3866, 2N3866A; $R_{\theta JA}$ shall be 325°C/W for 2N3866UB and 2N3866AUB.

4.5.3 Power output and collector efficiency measurements. The device shall be tested in the circuit of figure 4 using the procedure outlined on figure 5. The specified conditions shall be applied and the variable capacitors adjusted to obtain maximum power output. When the maximum power output is obtained, the collector current shall be measured and recorded. The collector efficiency shall be computed as follows:

$$\eta \text{ in percent} = \frac{P_O \text{ (watts)} \times 100}{28 \times I_C \text{ (amperes)}}$$

4.5.4 Burn-out by pulsing. The devices shall be tested in the circuit of figure 6. The voltage source shall be increased from zero until the specified current is reached. The current shall be maintained for the specified time.

* 4.5.5 Displacement damage characterization. For RHA devices, each supplier shall perform a displacement damage characterization. The characterization shall demonstrate exposure versus data and does not indicate pass or fail criteria. The exposure shall be conducted in accordance with method 1017 of MIL-STD-750. The following details shall apply:

a. Samples may be taken from any wafer of the qualification lot.

b. As a minimum, testing shall be at 2E+12 n/cm^2 plus two additional fluence levels chosen by the manufacturer.

c. If the device degrades less than 5 percent of the specification at the highest fluence level, a single data point may be sufficient.

Alternate package options may be substituted for characterization. The displacement damage characterization data shall be made available from the manufacturer.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0 (JAN and JANTX) n = 116 devices, c = 0 (JANTXV) n = 15 devices, c = 0 (JANS)				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Salt Atmosphere (Laser marked devices only) 4/	1041	n = 6 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 4/	1071	n = 22 devices, c = 0				
Fine leak Gross leak						
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs, n = 11 wires, c = 0				
Subgroup 2						
Thermal impedance 6/	3131	See 4.3.4.	$Z_{\theta JX}$			°C/W
Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 5 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	30		V dc
Collector-base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
Emitter-base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)EBO}$	3.5		V dc
Collector-emitter cutoff current	3041	Bias condition D; V_{CE} = 28 V dc	I_{CEO}		20	µA dc
Collector-emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES1}		100	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio 2N3866, 2N3866UB 2N3866A, 2N3866AUB	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1}	 15 25	 200 200	
Forward-current transfer ratio 2N3866, 2N3866UB 2N3866A, 2N3866AUB	3076	V_{CE} = 5.0 V dc; I_C = 360 mA dc; pulsed (see 4.5.1)	h_{FE2}	 5.0 8.0		
Collector-emitter saturated voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)}$		1.0	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES2}		2.0	mA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio 2N3866, 2N3866UB 2N3866A, 2N3866AUB	3076	V_{CE} = 5.0 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE3}	 7 12		
Subgroup 4						
Magnitude of small-signal short-circuit current transfer ratio 2N3866, 2N3866UB 2N3866A, 2N3866AUB	3306	V_{CE} = 15 V dc; I_C = 50 mA dc; f = 200 MHz	$\lvert h_{fe} \rvert$	 2.5 4.0	 8.0 7.5	
Open circuit output capacitance	3236	V_{CB} = 28 V dc; I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		3.5	pF
Power output		V_{CC} = 28 V dc; P_{in} = 0.15 W; f = 400 MHz (see figure 4 and 4.5.3)	P_{1out}	1.0	2.0	W
Power output		V_{CC} = 28 V dc; P_{in} = 0.075 W; f = 400 MHz (see figure 4 and 4.5.3)	P_{2out}	0.5		W

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Collector-efficiency		V_{CC} = 28 V dc; P_{in} = 0.15 W; f = 400 MHz (see 4.5.3)	$\eta 1$	45		%
Collector-efficiency		V_{CC} = 28 V dc; P_{in} = 0.075 W; f = 400 MHz (see 4.5.3)	$\eta 2$	40		%
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Collector-emitter breakdown voltage (clamped inductive)	3011	V_{BE} = -1.5 V dc; I_C = 40 mA dc (see figure 7)	$V_{(BR)CEX}$	55		V dc

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, steps 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroups 1 and 2.

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TABLE II. Group D inspection and end-point limits.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 5 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	30		V dc
Collector-base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
Emitter-base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)EBO}$	3.5		V dc
Collector-emitter cutoff current	3041	Bias condition D; V_{CB} = 28 V dc	I_{CEO}		40	µA dc
Collector-emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES1}		200	µA dc
Forward-current transfer ratio 2N3866, 2N3866UB 2N3866A, 2N3866AUB	3076	V_{CE} = 5 V dc, I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1} 5/	 [7.5] [12.5]	 200 200	
Forward-current transfer ratio 2N3866, 2N3866UB 2N3866A, 2N3866AUB	3076	V_{CE} = 5 V dc, I_C = 360 mA dc; pulsed (see 4.5.1)	h_{FE2} 5/	 [2.5] [4.0]		
Collector-emitter saturated voltage	3071	I_C = 100 mA dc, I_B = 10 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)}$		1.15	V dc

See footnotes at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2						
* Steady-state total dose irradiation	1019	Gamma exposure, Condition A, V_{ECS} = 24 V				
Collector-emitter breakdown voltage	3011	Bias condition D; I_C = 5 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	30		V dc
Collector-base breakdown voltage	3001	Bias condition D; I_C = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)CBO}$	60		V dc
Emitter-base breakdown voltage	3026	Bias condition D; I_E = 100 µA dc; pulsed (see 4.5.1)	$V_{(BR)EBO}$	3.5		V dc
Collector-emitter cutoff current	3041	Bias condition D; V_{CB} = 28 V dc	I_{CEO}		40	µA dc
Collector-emitter cutoff current	3041	Bias condition C; V_{CE} = 55 V dc	I_{CES1}		200	µA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1} 5/			
2N3866, 2N3866UB 2N3866A, 2N3866AUB				[7.5] [12.5]	200 200	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc, I_C = 360 mA dc pulsed (see 4.5.1)	h_{FE2} 5/			
2N3866, 2N3866UB 2N3866A, 2N3866AUB				[2.5] [4.0]		
Collector-emitter saturated voltage	3071	I_C = 100 mA dc, I_B = 10 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)}$		1.15	V dc

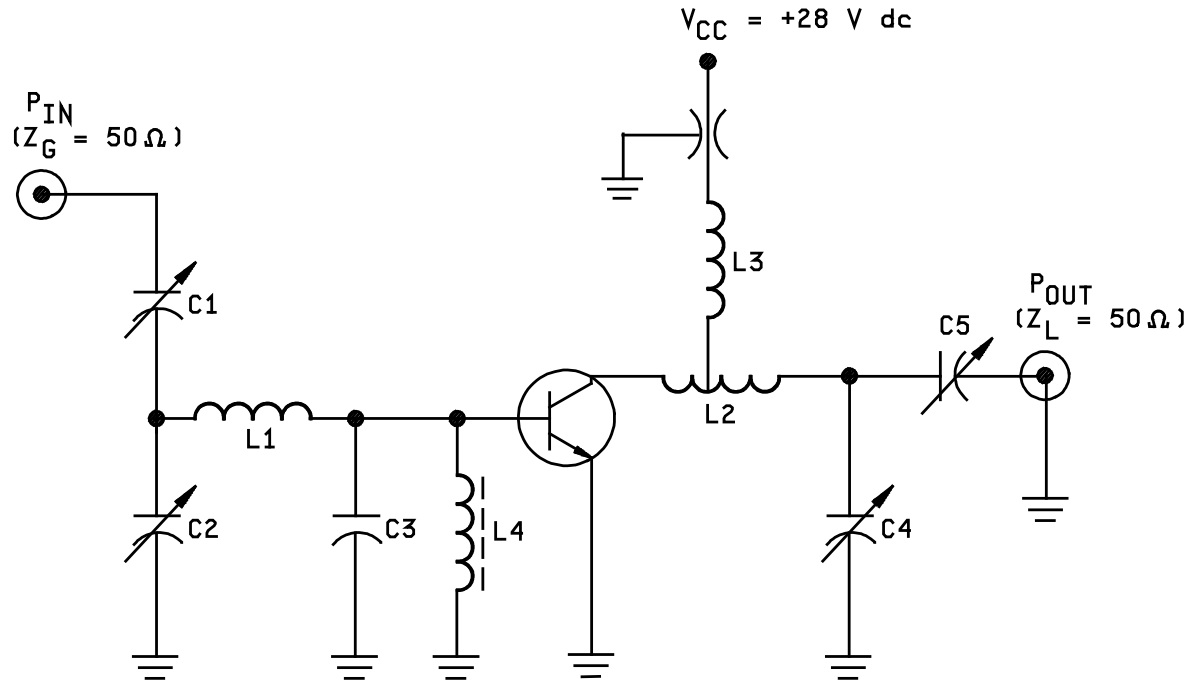
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding AL, UA, UB, and UBC suffix versions unless otherwise noted.
4/ See 6.2.e herein.
5/ See method 1019 of MIL-STD-750, for how to determine [h_{FE}] by first calculating the delta (1/h_{FE}) from the pre- and post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) – for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles, adjust power or current to achieve a ΔT_J = +100°C.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500, table E-IX, Subgroup 4	
Subgroup 5			
Not applicable			
Subgroup 6			
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	
Subgroup 12			
* Neutron Irradiation	1017	See 4.5.5	

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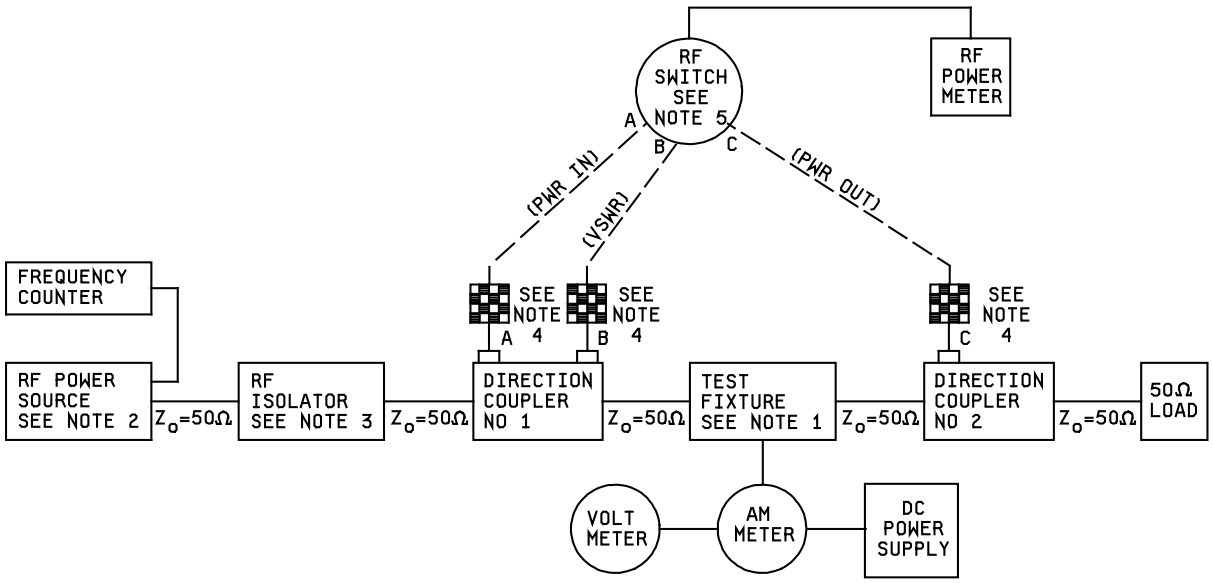


C_1, C_2, C_5 = 3 - 35 pF.
C_3 = 24 pF (see note).
C_4 = 0.4 - 7 pF.
L_1 = Straight piece number 16 bare tin wire, .625 inch (15.87 mm) long.
L_2 = 3 turns number 16 wire, .250 inch (6.35 mm) ID, .312 inch (7.92 mm) long.
L_3 = 1 turns number 18 wire, .250 inch (6.35 mm) ID, .022 inch (0.56 mm) long.
L_4 = Ferrite RF choke, Z = 450 Ω.

NOTE: For optimum performance, C_3 should be mounted as close as possible to base lead.

FIGURE 4. Power output test circuit (400 MHz).

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NOTES:

1. Test fixture is the circuit as described on figure 4.
2. RF power source may be any unit capable of generating desired power level at desired frequency with a harmonic and spurious content at least 20 dB below operating frequency level.
3. The RF isolator may be any device (pad, circulator, ect.) capable of establishing at least 20 dB of isolation (RL > 20 dB) between RF source and test fixture.
4. Variable attenuators (or fixed if calibrated): Attenuator on directional coupler number 2 shall be calculated against known working standard either by means of calibration chart or suitable adjustment if variable. Attenuator at position "A" of directional coupler number 1 shall be calibrated or adjusted so that actual power at test fixture is known. Attenuator at position "B" shall be adjusted to establish sensitivity needed to measure VSWR.
5. RF switch may be eliminated if additional power meters are used.

PROCEDURE:

a. Remove "test fixture" and install jumper between directional coupler number 1 and directional coupler number 2.
b. Set the RF switch to power output position "C".
c. Adjust frequency and power of RF source, as required by specification, and monitor frequency counter and RF power meter respectively (see note 4).
d. Set the RF switch to position "A" and adjust variable attenuator to obtain identical reading as power out in position "C" (see note 4).
e. Reconnect "test fixture" in test setup and insert device.
f. Adjust power supply to 28 V dc.
g. Adjust circuit output tuning for maximum power gain and circuit input tuning for maximum VSWR. (Switch between power in; VSWR, and power out while tuning and repeat as many times as necessary to obtain minimum VSWR and maximum power out. Check power in level before taking final reading. Minimum VSWR is defined as minimum reading obtained on power meter with switch in position "B" and maintaining power in.)

FIGURE 5. RF power output (P_{OUT}) test procedure.

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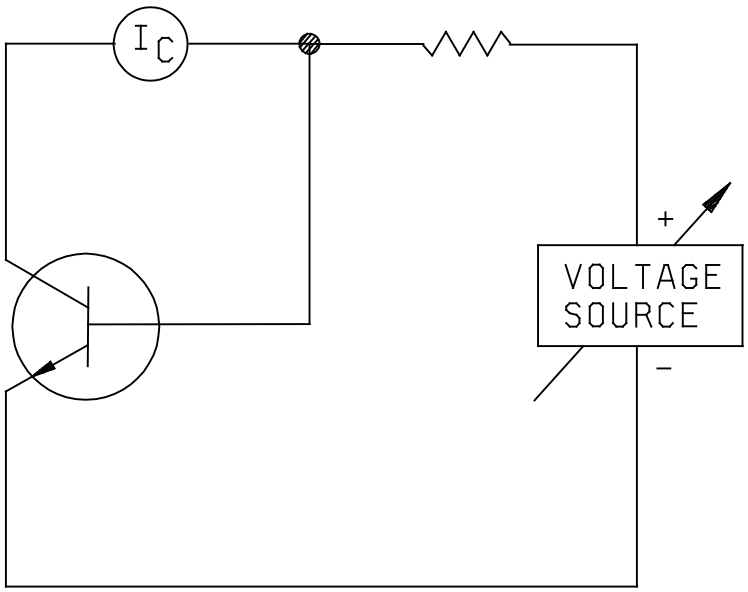
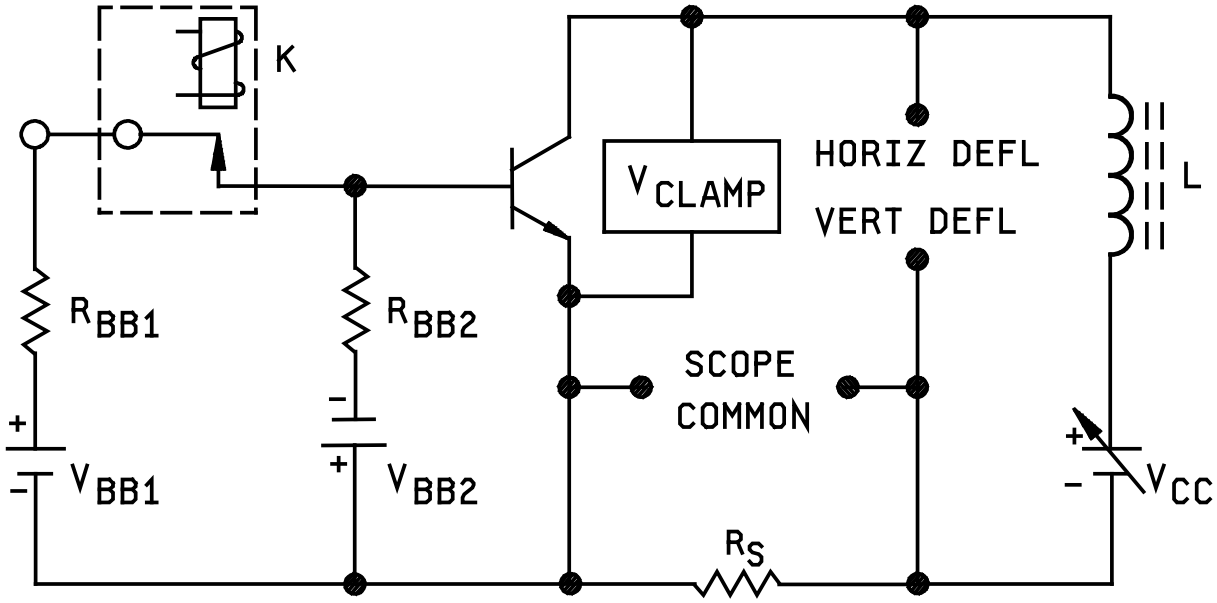


FIGURE 6. Burn-out by pulsing test circuit.

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R_{BB1} = 150 Ω.
V_{BB1} = 20 V dc.
K = s.p.s.t relay, 6 V ac coil (Clare Mercury Relay, model number HGP-1400, or equivalent).
R_{BB2} = 33 Ω.
V_{BB2} = 1.5 V dc.
R_S = 1 Ω ±1 percent, .5 watt (non-inductive).
V_{CC} = The voltage should be adjusted to approximately 17 volts.
L = 25 mH, 100 mA, 83 Ω resistive (Miller number 957, or equivalent).
Vclamp = 55 V (min).
$V_{(BR)CEX}$ clamped at 10 percent over rating.

FIGURE 7. $V_{BR(CEX)}$ (clamped inductive) test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN (see 1.5 and 6.5).

e. For acquisition of RHA designed devices, table II, subgroup 1 testing of group D is optional. If subgroup 1 testing is desired, it should be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N3866) will be identified on the QML.

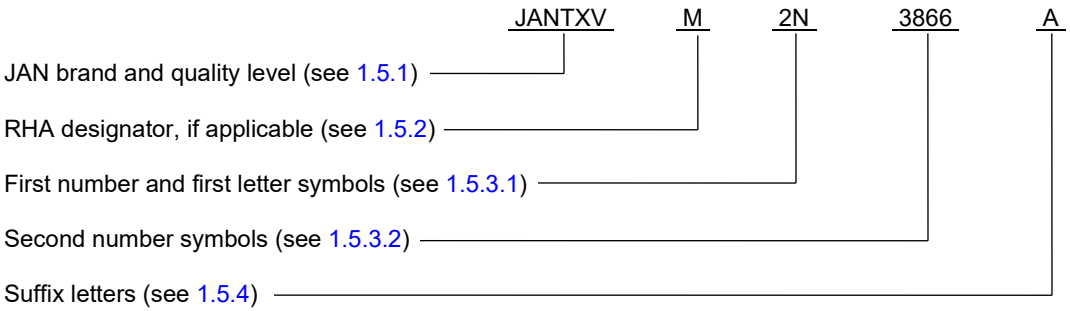
Die ordering information (1)	
PIN	Manufacturer
	34156
2N3866	JANHCA2N3866
2N3866A	JANHCA2N3866A

(1) For JANKC level, replace JANHC with JANKC.

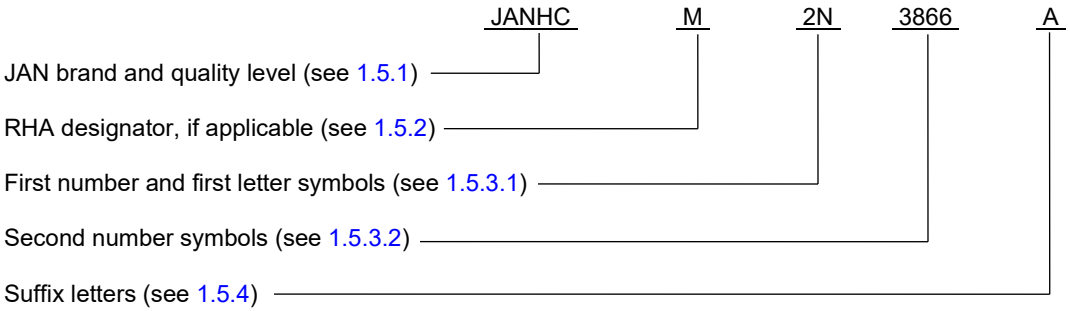
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6.5 PIN construction examples.

6.5.1 Encapsulated devices. The PIN for encapsulated devices are constructed using the following form:



6.5.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.6 List of PINs.

6.6.1 Encapsulated devices. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N3866	JANTX2N3866	JANTXV2N3866	JANS2N3866
JAN2N3866A	JANTX2N3866A	JANTXV2N3866A	JANS2N3866A
JAN2N3866UB	JANTX2N3866UB	JANTXV2N3866UB	JANS2N3866UB
JAN2N3866AUB	JANTX2N3866AUB	JANTXV2N3866AUB	JANS2N3866AUB

PINs for devices of the "TXV" quality level with RHA (1)	PINs for devices of the "S" quality level with RHA (1)
JANTXV#2N3866	JANS#2N3866
JANTXV#2N3866A	JANS#2N3866A
JANTXV#2N3866UB	JANS#2N3866UB
JANTXV#2N3866AUB	JANS#2N3866AUB

(1) The number sign (#) represents one of eight RHA designators available ("M", "D", "P", "L", "R", "F", "G", or "H").

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6.6.2 Unencapsulated devices. The following is a list of possible PINs available for unencapsulated devices covered by this specification sheet.

JANHCA#2N3866	JANHCA#2N3866A
JANKCA#2N3866	JANKCA#2N3866A

(1) The number sign (#) represents one of eight RHA designators available ("M", "D", "P", "L", "R", "F", "G", or "H").

6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS
* Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2023-037)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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